🫁 COMPREHENSIVE CDSS KNOWLEDGE BASE (PART 1)

SECTION 1: PNEUMONIA (COMMUNITY-ACQUIRED, HOSPITAL-ACQUIRED, & ATYPICAL)

Tier 1: Patient-Friendly Knowledge (For Empathy Narratives & Chatbot)

1. Comprehensive Definition & What is Happening to Your Body:

Pneumonia is a severe infection deep inside the lungs. Your lungs are made of microscopic air sacs called alveoli, which inflate like tiny balloons when you breathe to absorb oxygen. When pneumonia bacteria or viruses enter these sacs, your immune system rushes white blood cells to fight them. This battle causes the sacs to become inflamed and fill with fluid, pus, and cellular debris. Because the balloons are full of fluid, they cannot absorb oxygen, leaving you feeling exhausted and starved for air.

2. Detailed Symptom Timeline & Triggers:

- **Early Phase (Days 1-3):** Starts similarly to a cold or the flu. High fever, chills (rigors that make your teeth chatter), fatigue, and a dry cough.
- **Peak Phase (Days 4-7):** The cough becomes "productive," bringing up thick mucus (phlegm) that may be green, yellow, or rust-colored (tinged with blood). Breathing becomes rapid and shallow. Pleuritic chest pain occurs—a sharp, stabbing pain localized to one side of the back or chest that worsens when you take a deep breath or cough.
- **Recovery Phase (Weeks 2-4):** Fever breaks, but profound fatigue and a lingering dry cough can last for up to a month.

3. Home Care, Diet, & Precautions (Chatbot Advice):

- **Hydration is Medicine:** Drink at least 2 to 3 liters of water, warm tea, or broth daily. This physically thins out the thick mucus in your lungs, making it easier to cough up and expel.
- **Positioning:** Do not lie flat on your back. Prop yourself up with pillows at a 45-degree angle. This allows gravity to help your lungs expand fully and prevents fluid from pooling at the back of your lungs.
- **The "Finish the Pill" Rule:** You will likely feel 80% better after 3 days of antibiotics. **Do not stop taking them.** Stopping early leaves the strongest, most resistant bacteria alive, which will multiply and cause a second, much deadlier infection that normal antibiotics cannot cure.

4. High-Risk Scenarios (When to go to the Emergency Room):

- If your lips or fingernails turn a bluish or grayish color (cyanosis, meaning lack of oxygen).
- If you experience sudden confusion or disorientation (especially in elderly patients).
- If your breathing rate exceeds 30 breaths per minute while resting.

Tier 2: Advanced Clinical Protocols (For Radiologist/Physician Reports)

1. Pathophysiology & Etiology Breakdown:

- **Typical Bacterial CAP:** *Streptococcus pneumoniae* (most common, classic rust-colored sputum), *Haemophilus influenzae* (common in COPD/smokers), *Moraxella catarrhalis*, *Staphylococcus aureus* (often post-viral, e.g., after influenza, high risk of necrotizing pneumonia).
- **Atypical CAP:** *Mycoplasma pneumoniae* (classic "walking pneumonia" in young adults, bilateral diffuse infiltrates), *Legionella pneumophila* (associated with contaminated water sources, high fever, GI symptoms, hyponatremia), *Chlamydophila pneumoniae*.
- **Hospital-Acquired (HAP) / Ventilator-Associated (VAP):** *Pseudomonas aeruginosa*, MRSA (Methicillin-Resistant *Staph aureus*), *Klebsiella pneumoniae* ("currant jelly" sputum, common in chronic alcoholism).

2. Advanced Diagnostic Biomarkers:

- **Procalcitonin (PCT):** A crucial biomarker to differentiate bacterial from viral pneumonia. PCT levels >0.25 μg/L strongly encourage antibiotic initiation; <0.10 μg/L strongly discourage it.
- **C-Reactive Protein (CRP):** Used to monitor treatment response. A failure of CRP to drop by 50% within 4 days indicates treatment failure or a complicated effusion (empyema).

3. Comprehensive Severity Scoring (CRB-65 & PSI):

To determine patient disposition (Outpatient, Ward, or ICU), the system must evaluate the **CURB-65 score**:

- **C**onfusion (AMTS <= 8)
- **U**rea > 7 mmol/L (BUN > 19 mg/dL)
- **R**espiratory Rate $\ge$ 30 breaths/min
- **B**lood Pressure (Systolic <90 mmHg or Diastolic $\le$ 60 mmHg)
- **65:** Age $\ge$ 65 years.
- *Scoring:* 0-1 = Outpatient management. 2 = Hospital ward admission. 3-5 = Urgent ICU admission assessment.

4. Pharmacological Management & Dosages (Empiric Therapy):

- **Outpatient (Healthy, no recent antibiotics):** Amoxicillin 1g TID OR Doxycycline 100mg BID. (Macrolides like Azithromycin 500mg daily only if local pneumococcal resistance is $<25\%$).
- **Outpatient (Comorbidities/Heart Failure/Diabetes):** Amoxicillin/Clavulanate (Augmentin) 875mg/125mg BID + Azithromycin 500mg daily. Alternatively, monotherapy with respiratory fluoroquinolone (Levofloxacin 750mg daily).
- **Inpatient (Non-ICU):** Ceftriaxone 1g-2g IV daily + Azithromycin 500mg IV daily.
- **Inpatient (ICU):** Ceftriaxone 2g IV daily + Azithromycin 500mg IV OR Levofloxacin 750mg IV. If *Pseudomonas* is suspected, switch to Piperacillin/Tazobactam (Zosyn) 4.5g IV q6h. If MRSA is suspected, add Vancomycin (trough goal 15-20 mg/L) or Linezolid 600mg IV q12h.

SECTION 2: PLEURAL EFFUSION, EMPYEMA, & MALIGNANCY

Tier 1: Patient-Friendly Knowledge (For Empathy Narratives & Chatbot)

1. Comprehensive Definition:

Your lungs are wrapped in a thin, double-layered membrane called the pleura (like a lung inside a wet plastic bag). Usually, there are only a few teaspoons of lubricating fluid between these layers so your lungs glide smoothly when you breathe. A "pleural effusion" means liters of excess fluid have incorrectly pooled in this space. Because your ribcage is a rigid bone box, the extra fluid has nowhere to go but inward, physically crushing your lung and causing it to collapse under the water pressure.

2. What to Expect During a "Thoracentesis" (Pleural Tap):

- **The Procedure:** The doctor will have you lean forward on a table. They will use an ultrasound machine to find the deepest pocket of fluid. They will inject numbing medicine (lidocaine) into your back—this burns for about 10 seconds. Then, a small hollow needle is inserted between your ribs to vacuum out the fluid.
- **How it Feels:** You should not feel sharp pain during the draining, just a strange pulling or pressure sensation. As the fluid leaves and your lung suddenly pops back open, you may feel an overwhelming urge to cough. This is a good sign; it means the lung is re-expanding.
- **Post-Procedure:** You will have a small bandage on your back. Rest for 24 hours. A follow-up X-ray will be taken to ensure no air leaked into your chest during the procedure.

3. Understanding Fluid Types (Transudate vs. Exudate):

- **Watery Fluid (Transudate):** Usually caused by heart failure or kidney disease. The "plumbing" in your body is backed up, pushing clear, watery fluid into the lung cavity. The treatment is usually diuretic pills ("water pills") to make you urinate the excess fluid away.
- **Cloudy Fluid (Exudate):** Caused by inflammation, pneumonia, or cancer. The lung tissue is damaged, leaking thick, protein-rich fluid. This usually requires a physical chest drain and targeted treatments like antibiotics or chemotherapy.

Tier 2: Advanced Clinical Protocols (For Radiologist/Physician Reports)

1. Pathophysiology of Starling Forces:

Pleural effusions occur when the rate of fluid formation exceeds the rate of lymphatic removal.

- **Transudates** occur due to altered systemic hydrostatic or oncotic pressures (e.g., elevated left atrial pressure in Left Ventricular Heart Failure, or hypoalbuminemia in liver cirrhosis).
- **Exudates** occur due to increased capillary endothelial permeability caused by local inflammation (e.g., cytokine release in pneumonia) or impaired lymphatic drainage (e.g., tumor obstruction in malignancy).

2. Granular Fluid Analysis (Beyond Light's Criteria):

Once pleural fluid is aspirated, the following biochemical markers dictate the exact diagnosis:

- **Light's Criteria (Definitive Exudate):** Fluid/Serum Protein ratio >0.5, Fluid/Serum LDH ratio >0.6, or Fluid LDH $>2/3$ upper limit of normal serum LDH. (Note: If a patient is on diuretics for heart failure, a transudate may falsely appear as an exudate; calculate the Serum-Pleural Fluid Albumin Gradient. If gradient >1.2 g/dL, it is a transudate).
- **Glucose:** Pleural glucose <60 mg/dL indicates rheumatoid pleurisy, complicated parapneumonic effusion/empyema, malignant effusion, or tuberculous pleurisy.
- **Amylase:** Elevated pleural amylase (above upper limit for serum) strongly suggests acute pancreatitis, esophageal rupture (Boerhaave syndrome), or malignant effusion.
- **Cytology & Cell Count:** Lymphocyte predominance ($>50\%$) suggests tuberculosis or malignancy. Neutrophil predominance suggests acute parapneumonic infection or pulmonary embolism.

3. Empyema & Complicated Parapneumonic Effusion Management:

- **Stages of Empyema:** 1. Exudative (free-flowing fluid), 2. Fibrinopurulent (bacterial invasion, fibrin webs form causing loculations/pockets), 3. Organizing (fibroblast proliferation creating a thick "peel" over the lung, preventing expansion).
- **Mandatory Chest Tube (ICD) Indications:** Frank pus aspirated, positive Gram stain/culture of fluid, fluid pH <7.20, or fluid Glucose <3.4 mmol/L (<60 mg/dL).
- **Intrapleural Fibrinolytic Therapy (MIST-2 Trial Protocol):** For loculated/septated infected effusions that do not drain, instill tissue Plasminogen Activator (tPA) 10mg + Deoxyribonuclease (DNase) 5mg intrapleurally twice daily for 3 days to break down fibrin webs and reduce pus viscosity.

4. Malignant Pleural Effusion (MPE) Oncology Pathways:

- **Symptomatic MPE with Expandable Lung:** Perform chemical pleurodesis. Introduce sterile Talc slurry (usually 4g) via a chest tube to induce severe pleural inflammation, causing the visceral and parietal pleura to permanently scar and fuse together, obliterating the space so fluid cannot return.
- **Trapped Lung Syndrome:** If the tumor encases the lung preventing it from expanding to touch the chest wall, pleurodesis will fail. Management requires insertion of a tunneled Indwelling Pleural Catheter (IPC) (e.g., PleurX catheter) for home-based intermittent drainage by the patient or district nurse.

🫁 COMPREHENSIVE CDSS KNOWLEDGE BASE (PART 2)

SECTION 3: PNEUMOTHORAX (COLLAPSED LUNG)

Tier 1: Patient-Friendly Knowledge (For Empathy Narratives & Chatbot)

1. Comprehensive Definition & What is Happening:

A pneumothorax occurs when air escapes from the inside of your lung and becomes trapped in the pleural space (the cavity between your lung and your ribcage). Because your chest is a closed vault, this trapped air builds up pressure. This pressure pushes against the outside of your lung, squeezing it and causing it to deflate like a popped balloon.

2. Detailed Symptom Timeline & Triggers:

- **Spontaneous (No Trauma):** Often happens suddenly while resting or doing light activity. You will feel a sudden, sharp, tearing pain on one side of your chest. This is immediately followed by breathlessness.
- **Tension (Emergency):** If the air keeps leaking out of the lung but cannot escape the chest cavity, the pressure builds rapidly. Symptoms include extreme gasping for air, a racing heart, bulging neck veins, and a feeling of impending doom. This requires an immediate 911 call.

3. Home Care, Diet, & Strict Precautions (Chatbot Advice):

- **Absolutely No Flying or Diving:** You are strictly forbidden from flying in an airplane or going scuba diving until a doctor has confirmed via X-ray that your lung is fully healed. The pressure changes in these environments can cause the lung to instantly collapse again, which is fatal in the air or underwater.
- **Smoking Cessation:** If you smoke, you must stop immediately. Smoking degrades the lung tissue, creating weak "blebs" (blisters) on the lung surface that pop and cause a pneumothorax. Continuing to smoke increases your chance of a second lung collapse by over 50%.
- **Example Case (Patient Context):** "A 22-year-old tall, thin college student who vapes suddenly grabs the right side of his chest while sitting in class, experiencing sharp pain and an inability to catch his breath."

Tier 2: Advanced Clinical Protocols (For Radiologist/Physician Reports)

1. Pathophysiology & Classification:

- **Primary Spontaneous Pneumothorax (PSP):** Occurs in patients without clinically apparent lung disease. Typically caused by the rupture of subpleural apical blebs or bullae. Demographically common in tall, thin males aged 10-30 years.
- **Secondary Spontaneous Pneumothorax (SSP):** Occurs as a complication of underlying lung disease (e.g., COPD, Cystic Fibrosis, severe Asthma, or Tuberculosis cavity rupture). SSP is far more dangerous due to the patient's already compromised pulmonary reserve.
- **Tension Pneumothorax:** A life-threatening medical emergency where a "one-way valve" mechanism occurs. Air enters the pleural space during inspiration but cannot exit during expiration. Intrapleural pressure exceeds atmospheric pressure, causing mediastinal shift, compression of the contralateral healthy lung, and reduced venous return to the heart (resulting in obstructive shock and cardiac arrest).

2. Management Algorithms (BTS Guidelines):

- **Observation:** For small PSP (<2 cm visible rim of air on CXR) in asymptomatic patients, observe and discharge with written advice to return immediately if breathlessness worsens. Follow up in 2-4 weeks.
- **Needle Aspiration:** The first-line treatment for symptomatic PSP or a large PSP (>2 cm). Use a 14-16G cannula inserted at the 2nd intercostal space, mid-clavicular line. Aspirate up to 2.5 Liters. If successful, observe and discharge.
- **Intercostal Drain (ICD):** Mandatory for all SSPs, failed needle aspirations in PSP, or bilateral pneumothoraces. Insert a small-bore drain (8-14 French) in the "Safe Triangle" (bordered by the anterior border of the latissimus dorsi, the lateral border of the pectoralis major, a line superior to the horizontal level of the nipple, and an apex below the axilla).
- **Tension Protocol:** Do not wait for a chest X-ray if clinically suspected (tracheal deviation, hypotension, absent breath sounds). Immediately perform needle decompression at the 2nd intercostal space, mid-clavicular line, followed by formal ICD insertion.

SECTION 4: PULMONARY TUBERCULOSIS (TB) & CAVITARY DISEASE

Tier 1: Patient-Friendly Knowledge (For Empathy Narratives & Chatbot)

1. What is Tuberculosis?

TB is an infectious disease caused by a tough bacteria called *Mycobacterium tuberculosis*. Unlike normal pneumonia, TB bacteria can build a hard shell around themselves inside your lungs, allowing them to "sleep" (latent TB) for years. If your immune system weakens, they wake up (active TB) and slowly eat away at your lung tissue, creating holes (cavities).

2. Symptoms & Precautions:

- **Symptoms:** A chronic cough lasting more than 3 weeks, coughing up blood (hemoptysis), drenching night sweats (needing to change your bedsheets), unexplained weight loss, and extreme fatigue.
- **Precautions:** TB is highly contagious through the air. You must wear a mask and isolate yourself in a well-ventilated room until a doctor clears you. Treatment requires taking 4 different antibiotics at the same time for at least 6 months.

Tier 2: Advanced Clinical Protocols (For Radiologist/Physician Reports)

- **Radiological Presentation:** Reactivation (post-primary) TB typically presents with focal patchy opacities, volume loss, and cavitations in the apical and posterior segments of the upper lobes or the superior segments of the lower lobes.
- **Clinical Management:** Isolation in a negative-pressure room. Sputum Acid-Fast Bacilli (AFB) smear and GeneXpert MTB/RIF assay. Initiation of RIPE therapy: Rifampin, Isoniazid, Pyrazinamide, and Ethambutol for 2 months, followed by INH and RIF for 4 months.

SECTION 5: CONGESTIVE HEART FAILURE (CHF) & PULMONARY EDEMA

Tier 1: Patient-Friendly Knowledge

1. What is it? Sometimes, fluid in the lungs is not caused by an infection, but by a tired heart. If your heart is not pumping strongly enough, blood backs up into the blood vessels of your lungs. The pressure forces watery fluid out of the blood vessels and into the lung tissue (Pulmonary Edema).

2. Symptoms & Aftercare:

- **Symptoms:** Severe shortness of breath when lying flat in bed (needing to sleep propped up on 3 pillows), waking up gasping for air in the middle of the night, and swelling in your ankles and legs.
- **Precautions:** You must restrict your daily fluid intake and drastically cut down on salt (sodium). Salt acts like a sponge, holding excess water in your body and overloading your heart.

Tier 2: Advanced Clinical Protocols

- **Radiological Correlates:** Cardiomegaly (Cardiothoracic ratio >0.5), cephalization of pulmonary vessels, Kerley B lines (short, horizontal lines near the lung periphery indicating interstitial edema), and bilateral symmetric pleural effusions (transudative).
- **Clinical Management:** Assess NT-proBNP levels. Acute management includes LMNOP: Lasix (Furosemide IV), Morphine, Nitrates (vasodilation), Oxygen, and Positioning/Positive Airway Pressure (BiPAP).

SECTION 6: NORMAL CHEST X-RAY (BASELINE)

- **Clinical Assurance (Patient Facing):** "Your lungs appear completely clear. There are no signs of hidden fluid, active infections like pneumonia, or other structural abnormalities. Your heart size is normal. You can safely return to your normal routines."
- **Systematic Radiological Review (The 'ABCDE' Approach for Reports):**
    - **Airway:** Trachea is central and patent.
    - **Breathing:** Lungs are well-inflated bilaterally. No focal consolidations, masses, or pneumothorax.
    - **Circulation:** Heart size is within normal limits (CTR $<50\%$). Aortic arch and mediastinal contours are unremarkable.
    - **Diaphragm:** Right hemidiaphragm is slightly higher than the left (normal physiological finding due to the liver). Costophrenic and cardiophrenic angles are sharp and clear.
    - **Edges/Bones:** No obvious rib fractures or destructive bony lesions.

🧠 SECTION 7: RAG SYSTEM PROMPT LOGIC & REPORT GENERATION ALGORITHMS

(CRITICAL INSTRUCTIONS FOR AZURE OPENAI RAG ENGINE)

System Directive: You are a Dual-Tier Medical AI acting as the linguistic interface for a Clinical Decision Support System. You will receive an input containing: 1) The AI Vision Model's Predicted Pathology, 2) The Confidence Score, and 3) A description of the Grad-CAM++ Heatmap. You must use the Knowledge Base to generate exactly two outputs.

ALGORITHM 1: Generating the Tier 2 (Radiologist/Technical) Report

- **Rule 1.1 (Structured Format):** The report must follow standard radiology syntax: *Clinical Indication, Findings, and Impression.*
- **Rule 1.2 (Pneumonia Mapping):** If the prediction is Pneumonia and the heatmap shows "focal lower lobe opacities," the Impression must state: "Focal consolidation consistent with Community-Acquired Pneumonia." The system MUST generate a recommendation note stating: "Clinical correlation required to assess CURB-65 score; if respiratory rate >30 or hypotension is present, evaluate for severe CAP protocols as per IDSA guidelines."
- **Rule 1.3 (Pleural Effusion Mapping):** If the prediction is Pleural Effusion and the heatmap shows "dense pooling obscuring the costophrenic angles," the Impression must state: "Moderate-to-large pleural effusion." The system MUST generate a recommendation note stating: "Recommend diagnostic thoracentesis to differentiate

exudate vs. transudate via Light's criteria. Evaluate pleural fluid pH; if ≤7.20, urgent intercostal drain (ICD) is indicated per BTS guidelines."

- **Rule 1.4 (Pneumothorax Mapping):** If the prediction is Pneumothorax, explicitly measure the threat. Output: "Radiolucent zone devoid of lung markings." If mediastinal shift is detected in the input, output MUST flash: "**CRITICAL ALERT:** Radiological signs of Tension Pneumothorax. Recommend immediate needle decompression."

ALGORITHM 2: Generating the Tier 1 (Patient/Empathy) Narrative

- **Rule 2.1 (Tone):** The tone must be empathetic, calm, and written at an 8th-grade reading level. Do not use words like "exudate," "consolidation," or "thoracentesis" without explaining them simply.
- **Rule 2.2 (Structure):** The narrative must have four sections: 1. *Your Results*, 2. *What This Means*, 3. *Next Steps for Your Care*, 4. *Home Precautions.*
- **Rule 2.3 (Hallucination Constraint):** DO NOT prescribe specific antibiotic dosages to the patient. Tell them "Your doctor will prescribe a specific antibiotic." DO NOT diagnose cancer. If the doctor's report suspects Malignant Pleural Effusion, the patient report must simply state: "We have found fluid that needs to be tested by a specialist to understand its exact cause."

🫁 COMPREHENSIVE CDSS KNOWLEDGE BASE (PART 3)

SECTION 8: PEDIATRIC PNEUMONIA (WHO PROTOCOLS)

Tier 1: Patient-Friendly Knowledge (For Empathy Narratives & Chatbot)

1. Why is it different for children?

Pneumonia in babies and young children presents differently than in adults. It is the single largest infectious cause of death in children worldwide. Because their airways are very small, inflammation can block their breathing much faster than in an adult.

2. Emergency Warning Signs in Children:

- **Chest Indrawing:** When the child breathes in, the skin and muscles between their ribs or under their ribcage suck inward deeply.
- **Stridor:** A harsh, high-pitched whistling sound made when the child breathes *in* (even when calm and not crying).
- **Lethargy:** The child is unusually sleepy, difficult to wake up, or unable to drink fluids or breastfeed.
- If a parent notices any of these signs, they must take the child to the Emergency Room immediately.

Tier 2: Advanced Clinical Protocols (For Radiologist/Physician Reports)

1. Integrated Management of Childhood Illness (IMCI) Guidelines:

- The WHO promotes the IMCI approach for case management at all clinical levels.
- **Severe Pneumonia Classification:** Defined clinically by lower chest wall indrawing or the presence of general danger signs (inability to drink/feed, convulsions, lethargy, stridor in a calm child).
- **Fast Breathing Thresholds:** $\ge$ 60 breaths/min for children under 2 months; $\ge$ 50 breaths/min for children 2-11 months; $\ge$ 40 breaths/min for children 12-59 months.

2. Prevention Strategy Integration:

- Immunization against *Haemophilus influenzae type b* (Hib) and *Streptococcus pneumoniae* (pneumococcus) is the most effective preventative intervention.
- Malnutrition vastly increases mortality risk; clinical assessment must include nutritional status evaluation and exclusive breastfeeding history for infants under 6 months.

SECTION 9: ADVANCED DIAGNOSTICS & PATHOGEN RISK FACTORS (ATS/IDSA)

Tier 2: Advanced Clinical Protocols (For Radiologist/Physician Reports)

1. Pathogen-Specific Epidemiological Risk Factors:

When generating technical reports, the AI must cross-reference patient history with these specific risk profiles to suggest etiology:

- **Penicillin-Resistant pneumococcus (DRSP):** Age >65, alcoholism, immune-suppression, multiple medical comorbidities, or exposure to a child in a daycare center.
- **Enteric Gram-Negatives (*Klebsiella*, *E. coli*):** Residency in a nursing home, underlying cardiopulmonary disease, recent antibiotic therapy, or dysphagia/aspiration risk.
- **Pseudomonas aeruginosa:** Structural lung disease (e.g., bronchiectasis, severe COPD), chronic corticosteroid use (>10 mg prednisone/day), broad-spectrum antibiotic therapy for >7 days in the past month, or malnutrition.

2. Diagnostic Testing Criteria (When to test):

- **Blood Cultures:** While the yield is relatively low, blood cultures are strongly recommended for patients admitted to the ICU, patients with cavitary infiltrates, leukopenia, active alcohol abuse, or severe chronic liver disease.
- **Urinary Antigen Tests:** *Legionella* and *S. pneumoniae* urinary antigen tests are recommended for ICU patients, patients failing outpatient antibiotic therapy, or those with active alcohol abuse.

3. Strict Criteria for Clinical Stability (Switching to Oral Therapy):

A patient is not clinically stable and should not be discharged or switched to oral antibiotics until ALL of the following criteria are met:

- Temperature $\le$ 37.8°C (100°F)
- Heart rate $\le$ 100 beats/min
- Respiratory rate $\le$ 24 breaths/min
- Systolic blood pressure $\ge$ 90 mm Hg
- Arterial oxygen saturation $\ge$ 90% or pO2 $\ge$ 60 mm Hg on room air
- Ability to maintain oral intake and normal mental status.

SECTION 10: PROCEDURAL SAFETY & ULTRASOUND GUIDELINES (BTS)

Tier 2: Advanced Clinical Protocols (For Radiologist/Physician Reports)

1. The Role of Thoracic Ultrasound:

- Blind pleural fluid sampling (using only a chest X-ray for guidance) is obsolete and carries a high risk of iatrogenic pneumothorax or organ puncture.
- All pleural procedures for fluid (diagnostic aspiration, intercostal drain insertion) must be performed under real-time or site-marked **thoracic ultrasound guidance** by a trained operator .

2. Contraindications for Chest Drain Insertion:

- Uncorrected bleeding diathesis or coagulopathy (e.g., INR >1.5, Platelets $<50,000$).
- Overlying skin infection at the insertion site.

3. Complications to Monitor:

- **Re-expansion Pulmonary Edema (REPE):** Rapid drainage of a large volume of pleural fluid (>1.5 Liters at one time) or rapid evacuation of a large pneumothorax can cause the re-expanded lung to suddenly fill with edema fluid, leading to severe hypoxia and cardiovascular collapse.

🚨 SECTION 11: CRITICAL MEDICAL DISCLAIMER & LIMITATION OF LIABILITY

(MUST BE APPENDED TO ALL PATIENT-FACING AI OUTPUTS)

Disclaimer Template:

"Please note: This summary is generated by an Artificial Intelligence Clinical Decision Support System. It is designed solely for educational assistance and to help you understand your radiological findings. ***This is NOT a final medical report, and this AI cannot make definitive medical diagnoses.*** *The information provided here does not replace the professional judgment of a qualified healthcare provider. A mandatory consultation with your attending physician or radiologist is definitely needed to confirm these findings, discuss your specific medical history, and prescribe any actual treatments or medications. If you are experiencing severe shortness of breath, sudden chest pain, or confusion, please seek emergency medical attention immediately."*

🧠 SECTION 12: CHATBOT GUARDRAILS & EMERGENCY ESCALATION PROTOCOLS

(CRITICAL INSTRUCTIONS FOR AZURE OPENAI RAG ENGINE)

System Directive for Chatbot Interactions: You are an AI Medical Assistant interacting directly with a patient. You must adhere strictly to the following safety guardrails:

- **Rule 12.1 (Refusal of Direct Prescription):** If a patient asks "What antibiotic should I take?" or "What dose of medicine do I need?", you MUST REFUSE to answer.
    - *Action:* Output, "I cannot prescribe medication or specific dosages. Treatment for infections like pneumonia varies heavily based on your weight, allergies, and local resistance patterns. You must consult your doctor for a prescription."
- **Rule 12.2 (Refusal of Definitive Diagnosis):** If a patient uploads an image or describes symptoms and asks "Do I have cancer?" or "Do I have pneumonia?", you MUST NOT confirm a diagnosis.
    - *Action:* Output, "Based on the AI analysis of your X-ray, there are features consistent with [Pathology Name], but only a licensed doctor can officially diagnose you. Please review these results with your healthcare provider."
- **Rule 12.3 (Emergency Triage Protocol):** If the patient types ANY of the following keywords: *severe chest pain, coughing up large amounts of blood, cannot breathe, passing out, blue lips, severe confusion*, you MUST trigger the Emergency Protocol.
    - *Action:* STOP normal generation. Output in bold: **"🚨 URGENT: The symptoms you are describing are considered a medical emergency. Please stop using**

this chatbot and immediately call your local emergency services (e.g., 911) or go to the nearest hospital emergency room."

- **Rule 12.4 (Citation of the Disclaimer):** Every time you generate a summary based on a Grad-CAM++ heatmap, you must append the text from **Section 11 (Critical Medical Disclaimer)** at the very bottom of your response.